RISC-V / RUST / EMBEDDED ML

Scalar RISC-V. Faster INT8 inference.

1.59×

geometric-mean speedup across two models

Generated Rust kernels vs fastest tested C
ESP32-C3 · physical board measurements

riscv-int8-autotune

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Two real models.
Complete inference.

Person detector

376.3 → 259.2 ms **1.45×**

CIFAR-10 ResNet

620.4 → 354.7 ms **1.75×**

 Fastest of ESP-NN and two tested muRISCV-NN paths

riscv-int8-autotune

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Speed means little without correct outputs.

2,256

intermediate tensor checks

480

timed whole-model inferences

0

mismatches in three completed sessions

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Measured results. Open questions.

What changed

Blocking, packed weights, patch tiling, exact bias folding and bounded requantization.

What is public

Rust generator, reusable no_std kernels, raw logs, audits and reproducible charts.

What remains open

Intermittent QEMU stalls; broader model, board and power-cycle validation.